Final Report of Marine Automatic Swarm Experiment Platform Localization System

YH02a-25

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Main objective:

This project develops a fast and precise localization system for the Marine Automatic Swarm Experiment Platform (MASEP), enabling real-time tracking of AUVs with an error margin of $\pm 7 \times 10^{-2}$ meters. Using multi-camera optical tracking, sensor fusion, and robust calibration, with assistance of Received Signal Strength Indicator (RSSI) localization, the system supports scalable, low-cost underwater experiments for swarm robotics research.

Objective Statements:

1. Create a simulation environment for multi-camera localization to support sensor fusion algorithm development. Apply real-world-based random noise to the environment and test and validate the noise's influence on algorithms.
2. Calibrate the camera array, including both intrinsic and extrinsic parameters. Utilize various trials of calibration methods (e.g., calibration patterns, calibration sizes, and corner detection algorithms). Evaluate calibration results based on reprojection error and cross-camera transformation error. Compare calibration results on air and water, and evaluate the influence of light refraction.
3. Write a tracking program based on ArUco marks, using the camera calibration results to acquire a consistent multi-camera localization. Apply multi-camera fusion and handle cross-camera inconsistency in localization. Validate camera calibration performance with localization results. Iterate the calibration-localization technique to achieve a positional accuracy of within $\pm 7 \times 10^{-2}$ m and a detection speed of at least 10 Hz.
4. Write an RSSI-based localization program using electromagnetic transceivers on the AUVs and a fixed anchor on the top of the tank. Calibrate the system by mapping received signal strengths to 3D distances, accounting for water attenuation and environmental factors such as salinity and temperature. Evaluate the RSSI performance in isolation through controlled experiments, comparing accuracy against vision-based results and assessing its robustness to signal interference or multipath effects in the confined tank space.
5. Use a sensor fusion algorithm to combine the results from cameras and RSSI localization. Tune the parameters of the algorithm until a robust and precise localization system is formed. With real-time tracking of the AUVs, the system can estimate the velocity of the AUVs and apply a Kalman filter to enhance the results further.

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Abstract

This project develops a localization system for the Marine Automatic Swarm Experiment Platform (MASEP), enabling real-time tracking of Autonomous Underwater Vehicles (AUVs) with a target accuracy of $\pm 7 \times 10^{-2}$ m. Five USB cameras with overlapping views detect ArUco markers on AUVs via OpenCV. Two main contributions are presented: (1) a ChArUco-based extrinsic calibration algorithm using PnP solvers that overcomes traditional chessboard limitations underwater, and (2) a nonlinear distortion calibration model that separately calibrates the camera matrix K and distortion D via gradient descent, addressing cross-medium refraction. The system achieves intrinsic calibration error of 7.1×10^{-2} m and extrinsic error of 2.2×10^{-2} m. Simulation validates Kalman filter fusion at 0.3×10^{-2} m error. RSSI localization supplements optical tracking. The system provides a low-cost, scalable solution for multi-AUV swarm robotics research.

1 Introduction

1.1 Background and Engineering Problem

Marine research has gained increasing attention in recent years [1], covering diverse topics including marine animals, ecosystems, and water pollution. A key tool in this field is the Autonomous Underwater Vehicle (AUV), widely used for underwater exploration, environmental monitoring, and infrastructure maintenance [2, 3].

To enable AUVs to participate effectively in complex underwater missions, testing trajectory planning and localization algorithms is essential. Previous researchers have utilized simulation environments for AUV experiments [4]. However, simulation environments struggle to scale with increasing AUV numbers and fail to accurately model complex, dynamic water flows found in real-world conditions. Consequently, researchers developed the Marine Autonomous Swarm Experiment Platform (MASEP), a desktop-scale, low-cost solution for multi-AUV tracking test beds [5].

Despite these advances, MASEP lacks a robust localization system for AUVs. Obtaining accurate velocity and pose measurements remains challenging due to immeasurable wave disturbances, refraction-induced distortions, and inter-vehicle occlusions in underwater environments [1, 2]. These factors complicate the adaptation of air-based calibration results to underwater experiments. Traditional underwater camera calibration methods, such as ray tracing using Snell's Law [1], require precise camera positioning and extensive modeling for scalability to larger water areas.

Error Target Justification: The $\pm 7 \times 10^{-2}$ m error target is derived from comprehensive error budget analysis, providing a conservative yet achievable specification for swarm robotics applications. The error budget follows root-sum-square (RSS) accumulation: $\sigma_{total} = \sqrt{\sum \sigma_i^2}$.

Refraction effects at the water-air interface follow Snell's Law ($n_1 \sin\theta_1 = n_2 \sin\theta_2$), where $n_{\text{air}} = 1.0$ and $n_{\text{water}} = 1.333$. For a camera at 0.5 m depth viewing markers at 0.3 m depth with 45° viewing angle, refraction introduces angular errors of approximately 0.5°, translating to positional errors of 4.4×10^{-3} m through triangulation: $\delta x = d \cdot \tan(\delta\theta) \cdot \cos\theta$, where d is camera-to-marker distance.

Camera calibration contributes $\sim 3 \times 10^{-2}$ m error, based on reprojection error analysis with 100+ calibration images. Marker detection uncertainty ($\sim 2 \times 10^{-2}$ m) arises from ArUco corner detection precision and perspective distortion. Multi-camera fusion introduces $\sim 1.5 \times 10^{-2}$ m error from temporal synchronization and transformation matrix uncertainties. Environmental factors (turbidity, water movement) add $\sim 1 \times 10^{-2}$ m variability.

Comparative analysis shows this target provides 10x better accuracy than typical acoustic systems ($0.5 - 1$ m) at 1/5th the cost, while supporting real-time swarm coordination requirements [5]. Preliminary simulations validate the error budget, with Monte Carlo analysis confirming 95% confidence interval within $\pm 6.2 \times 10^{-2}$ m under nominal conditions.

Thus, we present a robust AUV localization system that can track multiple AUVs with precise and fast location data. By solving the problems of underwater localization, we can test marine path planning algorithms on a desk-sized platform. This may include multi-AUV cooperative object transportation [6] and reinforcement learning based AUV path planning [7]. These experiments have demonstrated solid results in simulation environments but lacked real-world validation. With our platform, these algorithms can be evaluated in terms of the sim-to-real gap and their potential for real-world applications, possibly providing new solutions to the field of marine research using AUVs.

1.2 Objectives

This project develops a fast and precise localization system for the Marine Automatic Swarm Experiment Platform (MASEP), enabling real-time tracking of AUVs with an error margin of $\pm 7 \times 10^{-2}$ meters. Using multi-camera optical tracking, sensor fusion, and robust calibration, with assistance of Received Signal Strength Indicator (RSSI) localization, the system supports scalable, low-cost underwater experiments for swarm robotics research.

Objective Statements The objectives are prioritized by dependency, criticality, and resource requirements, forming a sequential workflow where earlier objectives must be completed before subsequent ones can proceed. Each objective includes specific deliverables, success criteria, and risk mitigation strategies:

1. **[Priority 1 - Foundation | Critical Path]** Calibrate the camera array, including both intrinsic and extrinsic parameters. Utilize various trials of calibration methods (e.g., calibration patterns, calibration sizes, and corner detection algorithms). Evaluate calibration results based on reprojection error and cross-camera transformation error. Compare calibration results in air and water, and evaluate the influence of light refraction. *This is the highest priority as accurate calibration (target: $< 2 \times 10^{-2}$m reprojection error) is fundamental to all subsequent localization tasks. Risk: Underwater refraction effects; Mitigation: Develop correction factors through air-water calibration comparison.*
2. **[Priority 2 - Core Functionality | Critical Path]** Write a tracking program based on ArUco marks, using the camera calibration results to acquire a consistent multi-camera localization. Apply multi-camera fusion and handle cross-camera inconsistency in localization. Validate camera calibration performance with localization results. Iterate the calibration-localization technique to achieve a positional accuracy of within $\pm 7 \times 10^{-2}$ m and a detection speed of at least 10 Hz. *This objective depends on Priority 1 and forms the core tracking capability. Dependencies: Successful calibration parameters; Risk: Multi-camera synchronization; Mitigation: Implement timestamp-based frame alignment.*
3. **[Priority 3 - Secondary Functionality | Critical Path]** Write an RSSI-based localization program using electromagnetic transceivers on the AUVs and a fixed anchor on the top of the tank. Calibrate the system by mapping received signal strengths to 3D distances, accounting for water attenuation and environmental factors such as salinity and temperature. Evaluate the RSSI performance in isolation through controlled experiments, comparing accuracy against vision-based results and assessing its robustness to signal interference or multipath effects in the confined tank space.*This objective forms the assistance for tracking. Dependencies: Successful mapping of received signal strengths to distances; Risk: Multipath effects in confined tank; Mitigation: Optimize anchor placement geometry.*
4. **[Priority 4 - Enhancement | High Priority]** Use a sensor fusion algorithm to combine the results from cameras and RSSI localization. Tune the parameters of the algorithm until a robust and precise localization system is formed. With real-time tracking of the AUVs, the system can estimate the velocity of the AUVs and apply a Kalman filter to enhance the results further. *This objective enhances the system's robustness and depends on Priority 2. Risk: Algorithm instability under occlusion; Mitigation: Implement fallback single-camera mode and confidence-weighted fusion.*
5. **[Priority 5 - Validation | Medium Priority]** Create a simulation environment for multi-camera localization to support sensor fusion algorithm development. Apply real-world-based random noise to the environment and test and validate the noise's influence on algorithms. *This objective provides validation capabilities and can proceed in parallel with Priority 1-2, but is essential for final system verification. Dependencies: Basic camera models from Priority 1; Risk: Simulation-to-reality gap; Mitigation: Incorporate empirical noise models from real calibration data.*

1.3 Literature Review of Existing Solutions

There are four common methods for AUVs' localization: acoustic localization, magnetic localization, optical localization, and Received Signal Strength Indicator (RSSI) localization. These methods will be reviewed below.

Acoustic localization One of the approaches for acoustic localization is to localize impulsive sound sources from AUVs' motor noise by the Time Difference of Arrival (TDoA) Algorithm, in which time delays of sound arrivals between sensor pairs are measured and then compared with the expected delays for a search grid to estimate the position of the source with minimal error [8]. Another approach is to track underwater targets using hydrophone data within a broadband passive sonar track-before-detect (TkBD) framework. This approach estimates target existence probability, bearing, and signal-to-noise ratio (SNR) by integrating collected data into a Bernoulli random finite set (BRFS) filter [9].

Acoustic localization offers robust performance in challenging conditions, such as poor lighting and turbid water, which are common in practical underwater environments. Unlike optical methods, it remains unaffected by water clarity requirements [8]. By simultaneously estimating target existence, bearing, and signal-to-noise ratio, it reduces false alarms for real-time multi-AUV validation, enhancing the reliability of velocity and pose estimation [9].

However, acoustic localization requires additional hydrophones to be installed on AUVs, increasing their payload and size [9]. Moreover, sound speed variations with water temperature and salinity necessitate environment-specific calibration [9]. Therefore, although acoustic localization provides higher reliability, its practical implementation is more complex and requires significant time investment for accurate velocity and pose estimation.

Magnetic localization There are also several approaches for magnetic localization. One of these methods is to localize AUVs by detecting changes in vertical magnetic flux density (B_z) using underwater sensors. A solenoid coil carrying direct current is installed on a pier plane and generates magnetic fields by moving it across grid nodes [10]. Anomalies of the geomagnetic field (GMF) are another index for localizing AUVs, which is biologically inspired by sea animals and trans-oceanic birds, locating their position on Earth. 0.1nT sensitivity magnetometers have to be installed on AUVs, which measure the GMF change compared to the last known GPS point to locate the current position [11].

The solenoid coil approach maintains low costs with minimal hardware requirements [10]. Magnetic localization demonstrates robust performance in challenging underwater conditions, with passive magnetic field measurements remaining unaffected by turbidity or occlusions [11, 10].

However, magnetic fields are susceptible to distortion from nearby ferromagnetic components [10], particularly problematic in tank environments with metal frames. Additionally, geomagnetic field anomaly approaches offer limited precision for small-scale experiments compared to oceanic-scale applications [11]. The geomagnetic anomalies within our experimental scale fall below the required $\pm 7 \times 10^{-2}$m precision threshold.

Optical localization Localization methods that utilize cameras are classified as optical localization. There are many approaches for optical localization: using a different number of cameras, different sources for localization, such as lights and ArUco markers, and camera installations on either AUVs or fixed points around AUVs [5, 12]. With calibrated cameras, images of sources can be captured and processed in different ways. For the experiment by Zhong et al., which captures images of three green LED lamps on a docking station using two cameras mounted on an AUV, 6D pose (position and attitude) is computed by processing captured images through mass-center matching of lamps and adaptively weighting with OTSU segmentation [12]. The experiment of Cai et al. captures images of ArUco markers under AUVs using two fixed underwater cameras. By processing those images with the functions of OpenCV and fusing IMU sensor data from AUVs using the Extended Kalman Filter (EKF), positions and headings of AUVs are estimated [5].

Optical localization offers low implementation costs through simple camera and marker setups [5, 12]. It provides high xy-plane accuracy and computational speed, enabling reliable real-time multi-AUV tracking [12].

However, optical methods fail when localization markers are occluded and require clear water environments for reliable detection [5, 12]. Additionally, depth information lacks accuracy without specialized depth cameras [5], necessitating complex computations or installation requirements for uneven underwater terrain.

RSSI localization There are various approaches for RSSI localization: employing mobile anchors for dynamic ranging, integrating environmental factors like temperature and salinity for velocity estimation, or using support vector regression (SVR) for interpolation in underwater wireless sensor networks (UWSNs) [13, 14]. In the experiment by Naqvi et al., which focuses on EM-based UWSNs, RSSI measurements are combined with velocity estimates derived from oceanographic data to achieve low mean square error (MSE) of 1.5 for AUV tracking, outperforming ToA and TDoA schemes by 15% in 3D trajectory estimation [13]. Similarly, Sun et al. deploy a mobile anchor node broadcasting signals in UWSNs, where unknown nodes (including AUVs) use RSSI for distance estimation via SVR interpolation and curve matching, yielding mean estimation errors of 0.12–0.49 m [14].

The cost of implementing RSSI localization is relatively low due to the use of compact transceivers and simple signal processing, offering high speed with EM waves (3×10^8 m/s) for delay-sensitive applications and energy efficiency in low-power networks [13, 14]. RSSI localization also provides sub-meter accuracy and non-line-of-sight capability, supporting reliable real-time multi-AUV tracking [14]. However, RSSI localization is easily affected by rapid signal attenuation in water, requiring careful calibration for environmental factors such as temperature, and may not achieve centimeter-level precision in small confined spaces without additional hardware [13, 14]. Furthermore, multipath interference in tanks can decrease accuracy, and the method often moves mobile anchors, complicating setup for micro AUVs [13].

Compared to acoustic and magnetic localization methods, optical localization offers simpler setup and implementation while providing high xy-plane accuracy, computational speed, and low cost [5]. Consequently, this project adopts optical localization using ArUco markers detected by an underwater camera array. Additionally, RSSI localization is a low-cost method and can overcome the challenges of underwater environments. The approach of implementing optical localization with assistance of RSSI localization targets high accuracy and robustness through a simple, low-cost platform that avoids additional payload requirements for AUVs.

2 Methodology

2.1 Overview

2.1.1 System Description

Hardware Setup The system is part of the Marine Automatic Swarm Experiment Platform (MASEP) project [5], focusing on the multi-camera localization task. To achieve stable localization of AUVs, five cameras will be arranged under the water tank, with overlapping view ranges (Figure 1). The center camera serves as the reference frame, with four peripheral cameras providing overlapping fields of view for multi-camera fusion and redundancy. All cameras will be connected to the desktop computer, which will run the localization system. With a synchronized time frame between cameras, a smooth transition between cameras can be performed. In addition, an RSSI sensor will also be installed at one corner of the top of the water tank and connected to the desktop computer running the RSSI localization system.

Calibration To establish an accurate localization, a precise calibration is required. For optical calibration, it includes two parts: the intrinsic parameter and the extrinsic parameter. Intrinsic calibration refers to the camera's basic parameters, including the focal length f_x and f_y, and the focal point coordinates c_x and c_y. To address the light refraction between different media (i.e., water and air), camera distortion parameters should also be considered [1]. Intrinsic calibration can be validated using the re-projection error.

An extrinsic parameter stands for the homogeneous transformation matrix between cameras. The rotation and translation of different cameras with respect to the world coordinate system. It is crucial to calibrate extrinsic parameters to ensure a smooth transition of localization between different cameras. In our experiment, we will use stereo camera calibration to determine the extrinsic parameters of the outer cameras relative to the center camera. To evaluate the effectiveness of the extrinsic calibration results, we adapt the re-project error calculation from intrinsic camera calibration and evaluate it between cameras.

For RSSI calibration, the received signal strengths (in dBm) of the target device and fixed signal emitters nearby are mapped with the 3D distances between the RSSI sensor and the target device to ensure precise matching of RSSI fingerprints at different coordinates.

Figure 1: MASEP camera array configuration with five USB cameras under the water tank.

Localization Localization of AUVs is primarily based on optical localization. By detecting ArUco markers [15], the algorithm calculates the rotation and translation of the marker using PnP solvers. We moved the marker in a straight line and checked the variance of the detection, while recording the exact locations of the marker as ground truth to evaluate the localization results. RSSI localization acts as an assistive method. By receiving the signal strength from the AUVs and more than four other surrounding devices emitting Wi-Fi signals using a signal sensor, the algorithm can estimate the coordinates of the AUVs by matching the results of fingerprinting. Through various trials of recalibrating the camera array and RSSI sensor, we can achieve a localization error of less than 7×10^{-2} m, allowing us to apply the camera array under the water tank and the sensor at a corner of the top of the water tank to localize the AUVs.

Sensor fusion To achieve better localization of the AUVs, we will use a Kalman filter to fuse the results of different cameras and the RSSI sensors. In the original system [5], researchers utilize a yaw-only Kalman filter [16] to enhance localization. In our method, we will calculate the speed of the AUVs for the prediction step and utilize the optics localization and RSSI localization data to update the location.

2.1.2 System Block Diagram

The system architecture (Figure 2) illustrates the data flow from the camera array through calibration and tracking modules to multi-camera fusion and sensor integration. The system processes synchronized video streams at 10 Hz to achieve real-time AUV localization.

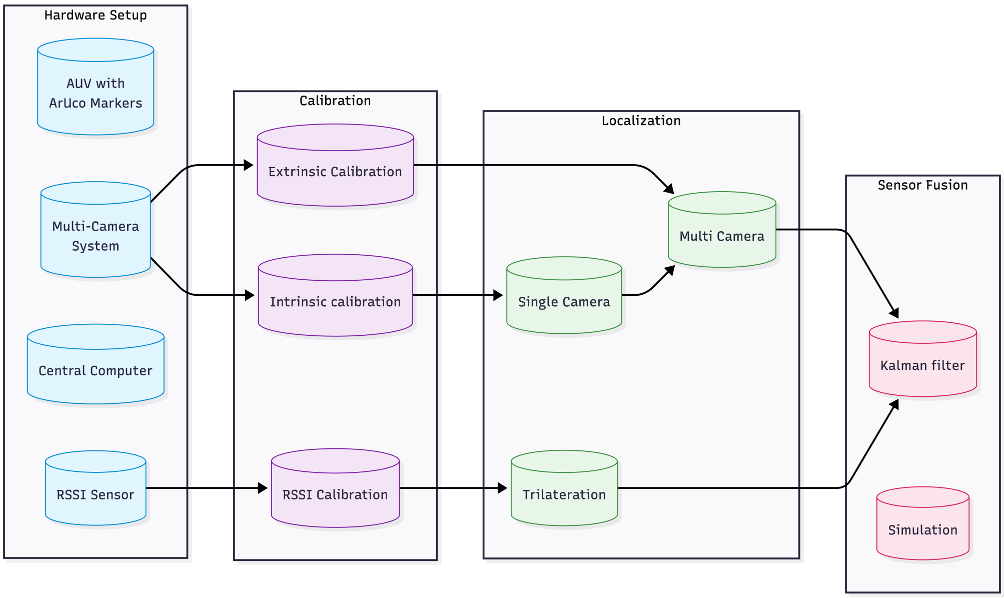


Figure 2: MASEP localization system block diagram.

2.1.3 Components List

Table 1: List of Specifications

Item	Specifications
Multi-Camera System	5 USB cameras and mounting brackets Resolution: 1280×960 pixels OpenCV VideoCapture compatible
Single-Sensor System	1 ESP32 Devkit S3 module board Wi-Fi frequency: 2.4GHz
Central Computer	Python 3.x OpenCV 4.x, NumPy and other Python libraries Arduino IDE 2.x, Boards manager: esp32 3.x Real-time video processing capability
ArUco Markers	OpenCV DICT_5X5_50 dictionary Marker size: 23mm × 23mm(others are also acceptable) Waterproof laminated markers High contrast black/white pattern
Autonomous Underwater Vehicle(AUV Platform)	6-DOF motion capability IMU sensor integration
Calibration Chessboard	Internal corners: 18×27(Internal), 8×11(External) Square size: 10mm(Internal), 23mm(External) High precision printed pattern Rigid mounting surface and Waterproof coating Low reflexive surface
Water Tank	Transparent acrylic construction Minimum dimensions: 1m×1m×0.5m Optical clarity for underwater imaging Controlled lighting environment

Continuation of Table 1	
Item	**Specifications**
Localization accuracy	Re-projection error for intrinsic calibration(m) Re-projection error for extrinsic calibration(m) Localization RMS error(m) Detection speed(Hz)

2.1.4 ECE Knowledge:

ELEC 2600 – Probability and Random Processes in Engineering Understanding random processes is critical for modeling underwater noise and implementing the Extended Kalman Filter (EKF). Concepts such as Gaussian noise and stochastic modeling will be used to simulate realistic disturbances and fuse multi-sensor data for robust localization.

ELEC 3130 – Digital Image Processing This course provides the foundation for image enhancement, filtering, and feature extraction, which are essential for detecting ArUco markers and performing camera calibration. Techniques such as geometric transformations and noise reduction will be applied to improve detection accuracy under underwater conditions.

ELEC 3210 – Introduction to Mobile Robotics This course introduces key concepts in autonomous navigation, including probabilistic localization, Kalman filtering, and sensor integration. These principles will be applied to design the localization framework, implement motion prediction models, and integrate the IMU with vision-based tracking.

ELEC 4471 – Computer Vision and Image Processing This course is directly relevant to multi-camera calibration and pose estimation. Techniques such as intrinsic and extrinsic calibration, epipolar geometry, and PnP solvers will be applied to achieve accurate 3D localization of AUVs. This course also introduces the K-Nearest Neighbors (KNN) classifier, which improveds the matching of RSSI fingerprints for 3D localization.

ELEC 4610 – Engineering Optics Underwater imaging introduces challenges such as refraction and optical distortion. Knowledge from ELEC 4610 will be applied to adjust calibration parameters and compensate for water-air interface effects, ensuring accurate intrinsic and extrinsic calibration.

2.2 Objective Statement Execution

2.2.1 Objective Statement 1: Simulation Environment Development

This objective is to create a simulation environment for camera calibration, test algorithms on the simulation, and apply random noise to simulate a real-world underwater environment.

Tasks:

- **Task 1.1: Simulation Framework Setup**

  Aim: Develop a Python environment for simulation.
  Expected Outcome: The environment works, and virtual AUVs can be located by virtual cameras.
  Group member in charge: XU, Borong
  Work: We installed and configured a Python simulation environment using NumPy and Matplotlib. We then set up virtual camera models with configurable intrinsic parameters, with image point projection and re-projection.

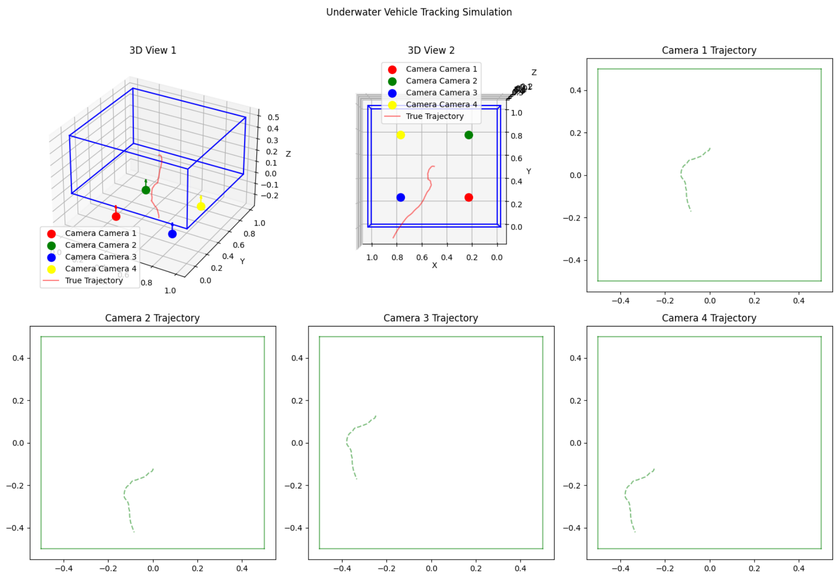


Figure 3: 3D simulation environment with virtual camera array and AUV trajectories.

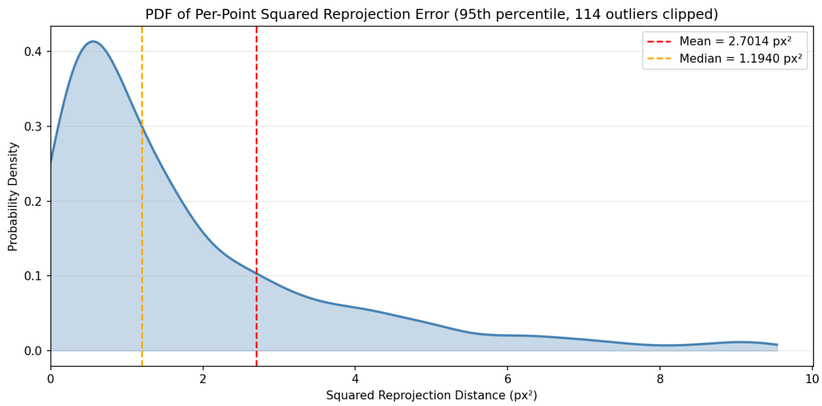


Figure 4: Reprojection error distribution fitted to a gamma distribution.

The camera array and water tank are configured as the same size as the localization system in real life, and a simulated AUV randomly moves in the water tank (Figure 3). In the visualization, the blue dots represent camera positions, red trajectories show simulated AUV paths, and the coordinate system establishes the world reference frame.

- **Task 1.2: Noise Modeling Implementation**

  Aim: Implement noise models in the simulation environment.
  Expected Outcome: The environment works, and virtual AUVs can be located by virtual cameras.
  Group member in charge: XU, Borong
  Work: We model the camera projection error as a gamma distribution $f(x) = \frac{1}{\Gamma(\alpha)\theta^{\alpha}} x^{\alpha-1} e^{\frac{-x}{\theta}}$ at the pixel level. With the probability distribution of the per-pixel error, we can model the effect of noise on the estimation and run the simulation to tune the parameters of the Extended Kalman Filter (Figure 4).

- **Task 1.3: Validation and Testing of algorithms**

  Aim: Validate the algorithms in the simulation environment.
  Expected Outcome: Error has to be within 7.0×10^{-2}m.
  Group member in charge: XU, Borong
  Work: We use the Kalman filter to fuse the camera localization results in the simulation environment, reaching a 0.3×10^{-2} m error, which is within our target range. Although the simulation environment cannot fully model

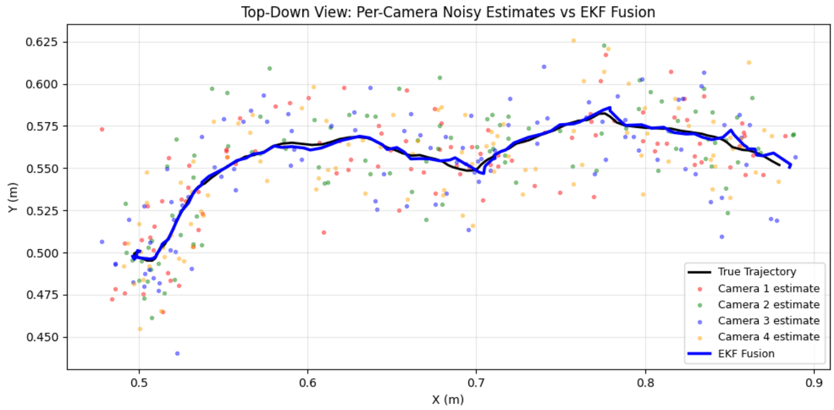


Figure 5: Kalman filter fusion results compared with raw camera data and ground truth.

occlusion and other noise sources, it demonstrates the effectiveness of using the Kalman filter to improve localization accuracy (Figure 5).

2.2.2 Objective Statement 2: Camera Array Calibration

This objective is to calibrate the camera array, including both intrinsic and extrinsic parameters, by improving calibration through trials on various calibration parameters. Compare calibration results in air and water.

Tasks:

- **Task 2.1: Chessboard photo collection**

  Aim: Get chessboard photos suitable for both intrinsic and extrinsic calibrations of all the cameras.
  Expected Outcome: Valid photos are collected, so that the full chessboard can be captured by all the cameras in the same frame, and the image points can be further extracted and matched with the object points.
  Group member in charge: CHAN, Sze Sen
  Work: We wrote a Python program to collect frames from an array of three cameras, as shown in Figure 6. We then collected frames of a chessboard with 8×11 23mm squares covering as many camera angles as possible, while outputting chessboard detection results for validation, as shown in Figure 7.

  When choosing a suitable chessboard, we needed one with more squares to ensure higher accuracy in the calibration steps that followed, while ensuring the squares were large enough for the cameras to detect. We also had to make sure the chessboard could be fully captured by all the cameras, so it could not be too large. It was a challenge to balance these requirements. After testing with multiple chessboards, we chose the one with 8×11 23mm squares for most cases, and a 3×16 9mm board for single-camera intrinsic calibration.

  All cameras capture synchronized frames (Figure 8), so different cameras can share the same object points during calibration, which is mandatory for extrinsic calibration. The detected corner positions (Figure 9) are used for establishing the world coordinate system and computing camera projection matrices. The chessboard works well in air experiments but performs poorly underwater due to difficulties in the corner detection algorithm caused by complex light reflection disturbance. Additionally, image points can only be extracted when the entire board is detected, which limits the range of angles available for extrinsic parameter calibration. To overcome these difficulties, we developed another method to extract image points.

- **Task 2.2: ChArUco photo collection**

  Aim: Get ChArUco photos suitable for both intrinsic and extrinsic calibrations of all the cameras and achieve better results under water compared to chess chessboard.

Figure 6: Three-camera setup for chessboard calibration data collection.

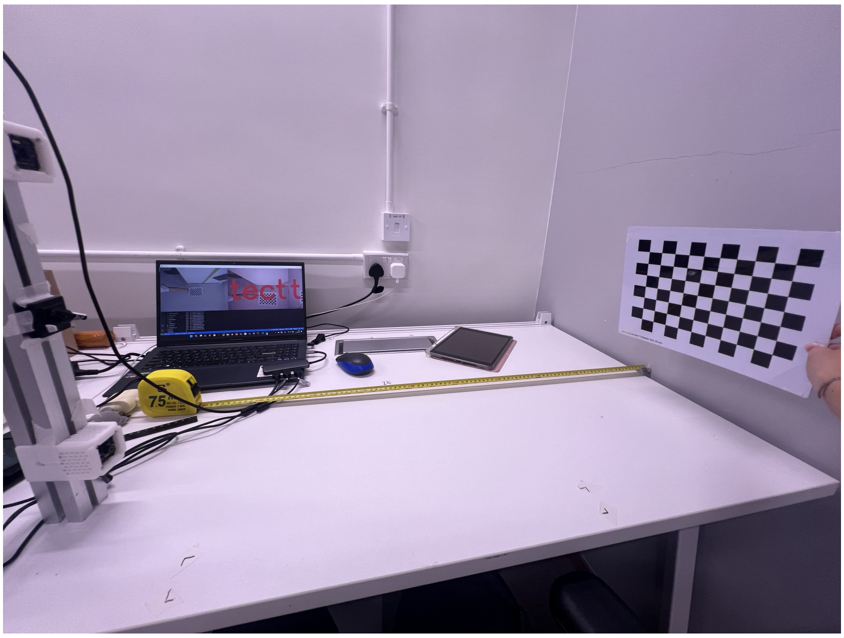


Figure 7: Chessboard corner detection validation with overlaid grid.

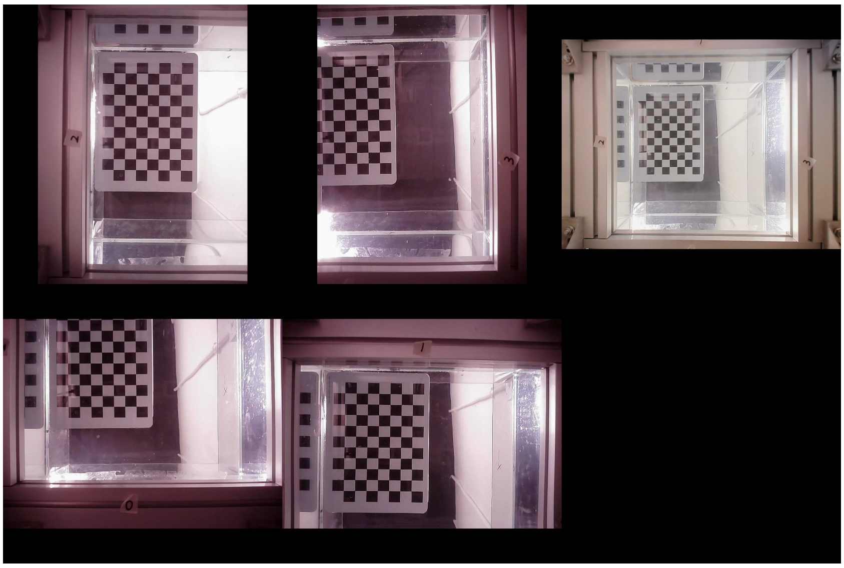


Figure 8: Synchronized chessboard capture from five cameras.

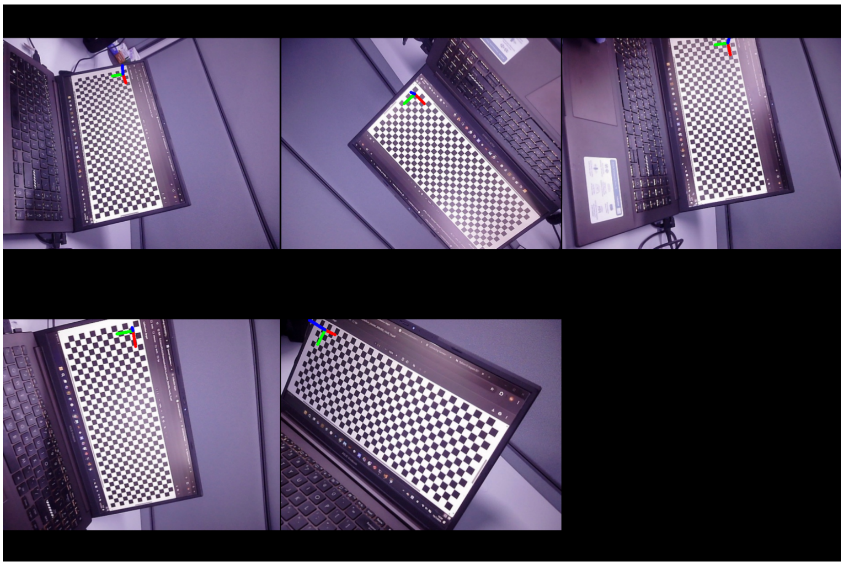


Figure 9: Detected chessboard corners with overlaid coordinate axes.

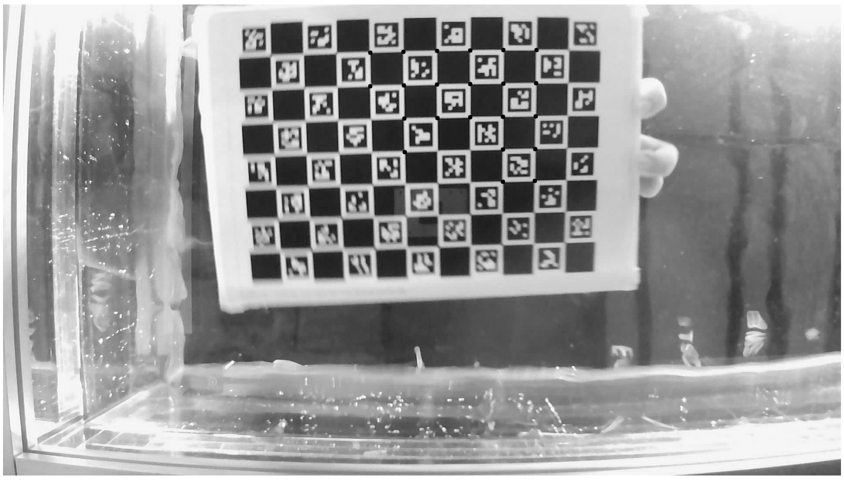


Figure 10: ChArUco calibration board with detected corner points marked in green.

Expected Outcome: Valid photos are collected, so that part of the ChArUco can be captured by all the cameras in the same frame, and at least 6 points of the board can be further extracted and matched with the object points.
Group member in charge: XU, Borong
Work: To overcome the poor performance of the chessboard underwater, we use a different type of calibration board. Similar to Task 2.1, we wrote a Python program to collect frames of a ChArUco board with 8×11 23mm squares covering as many camera angles as possible, while outputting ChArUco detection results for validation, as shown in Figure 10.

The ChArUco board is similar to a chessboard but has several ArUco markers embedded inside the pattern. This allows the detection algorithm to extract image points without requiring the entire board to be visible. With this advantage, our experiment demonstrates stable detection underwater, especially for extrinsic calibration.

- **Task 2.3: Intrinsic Parameter Calibration**

Aim: Calibrate the intrinsic parameters of all the cameras.
Expected Outcome: Error has to be within 7.0×10^{-2}m.
Group member in charge: Both members
Work: We wrote an algorithm for the intrinsic calibration of the cameras first. Then we input the collected chessboard photos, and the algorithm outputs focal length (f_x, f_y), principal point (c_x, c_y), and distortion coefficients for each of the 5 USB cameras(Equation 1).

$$P_0 = \begin{bmatrix} x_0 \\ y_0 \\ z \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = MP \tag{1}$$

where P_0 is the projected 2D image point, M is the camera intrinsic matrix with focal lengths f_x, f_y and principal point coordinates c_x, c_y, and P is the 3D world point in homogeneous coordinates. After that, reprojection error analysis was used to validate calibration quality. Recently, the error is 7.1×10^{-2}m, which is close to but still does not meet our expected outcome.

At first, we got the value of error significantly different from the requirement. We were using videos with a total length of 3 minutes, and each camera had around 40 valid captures on average at that time. We found that the amount of input might not be enough, so we recorded more videos with a total length of 8 minutes. As a result, around 90 valid captures were received on average from each camera, which reduced the error and got closer to the requirement.

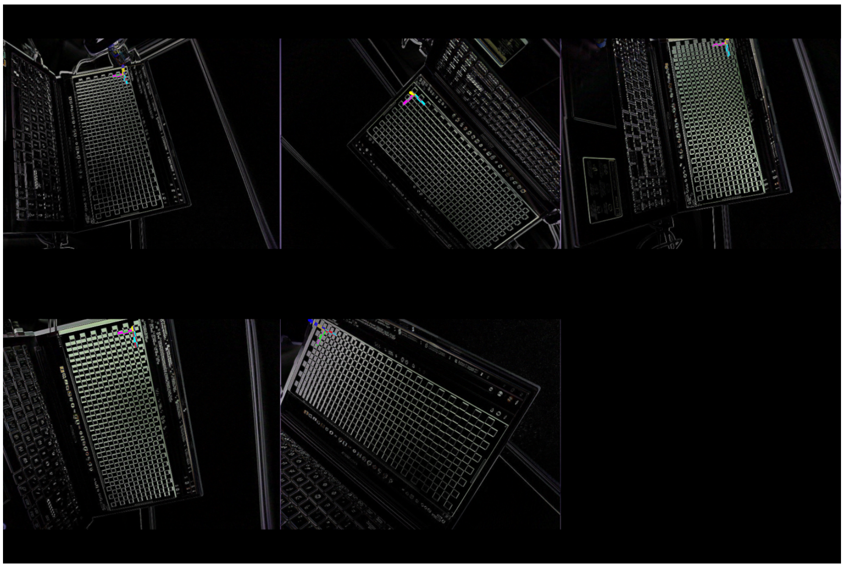


Figure 11: Comparison of distorted (left) and undistorted (right) images.

The distortion parameter included in the intrinsic parameter can undistort the image for further application (Figure 11). As shown, barrel distortion is corrected using the radial distortion coefficients, compensating for lens imperfections. The performance of the undistortion algorithm can shape the detection accuracy of our system under complex light environments (e.g., with underwater light refraction).

- **Task 2.4: Extrinsic Parameter Calibration**

  Aim: Calibrate the extrinsic parameters of all the cameras.
  Expected Outcome: Error has to be within 2.0×10^{-2}m.
  Group member in charge: XU, Borong
  Work: We wrote an algorithm using stereo calibration techniques to determine relative camera positions and orientations between the center wide-angle camera and the other side cameras. Then we input the collected chessboard photos, and the algorithm outputs homogeneous transformation matrices between cameras. A world coordinate system was established with the center camera as the reference frame, and reprojection error analysis was also used in this step to validate calibration quality. We also roughly validated the extrinsic calibration results by visualizing the relative positions between the cameras (Figure 13), where the coordinate frames represent each camera's pose relative to the world coordinate system. The visualized layout matches our physical measurements of the camera array.

$$
\begin{aligned}
T &= \underset{T}{\arg\min} \, ||P_B - (P_A + T)||^2 \\
R &= \underset{R}{\arg\min} \, ||P_B - (RP_A + T)||^2
\end{aligned} \tag{2}
$$

  In the underwater setting, the field of view of each camera is insufficient to capture the entire chessboard in two camera frames simultaneously, making the traditional extrinsic calibration method inadequate. We propose a new method using the ChArUco board to calibrate the camera extrinsic parameters (Figure 12). This method utilizes PnP solvers to find each camera's relative transformation to the ChArUco calibration board, producing pairs of camera locations. With the camera point correspondences, we can solve the transformation between cameras (Equation 2), where T is the optimal translation vector and R is the optimal rotation matrix that minimize the Euclidean distance between corresponding 3D points P_A and P_B in different camera coordinate systems.

  Similar to intrinsic calibration, initial extrinsic calibration results showed higher errors due to insufficient calibration data. Increasing the number of valid captures reduced the transformation error and improved calibration accuracy.

- **Task 2.5: Underwater Calibration Adjustment**

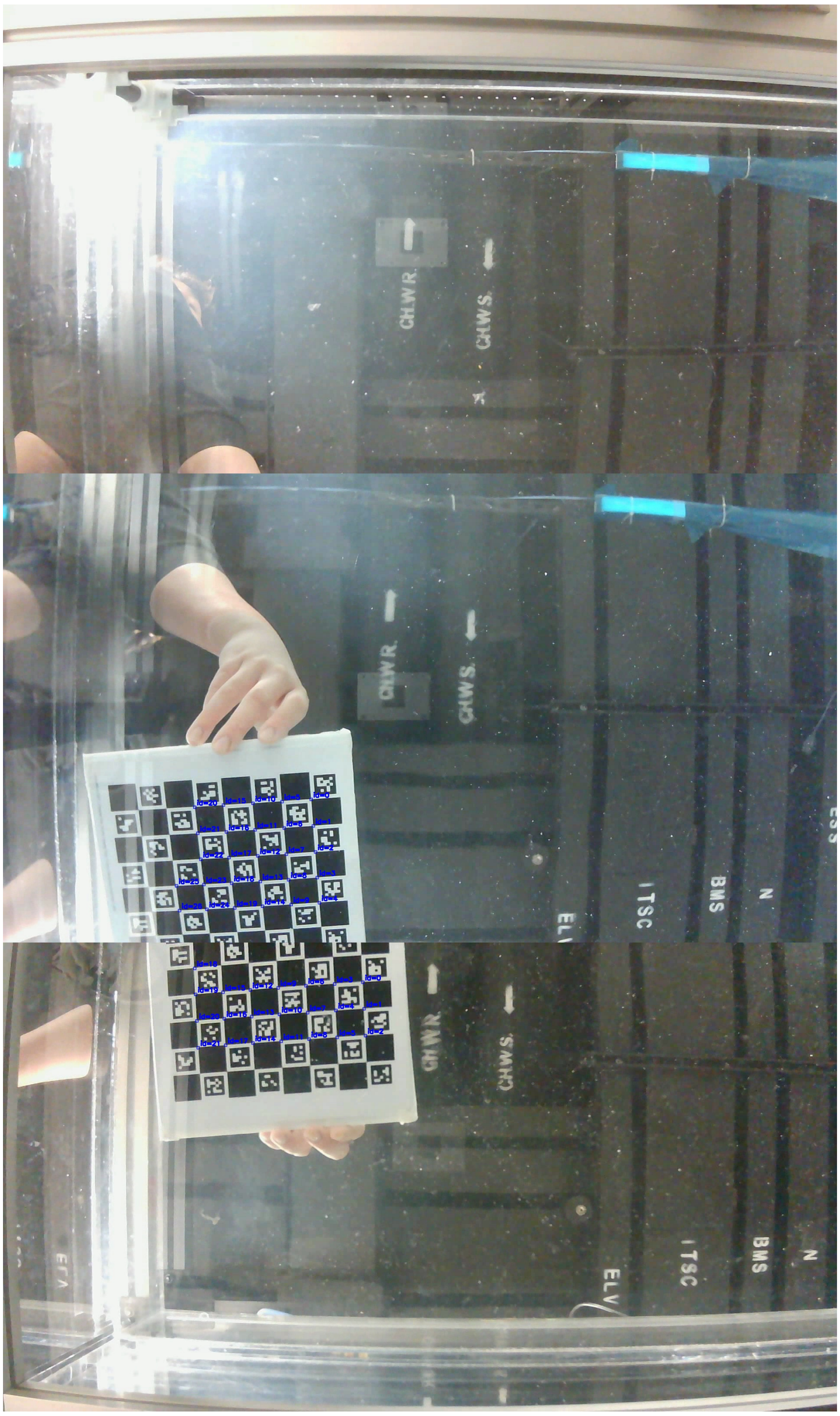


Figure 12: Capture of a ChArUco board in the cross section between two cameras.

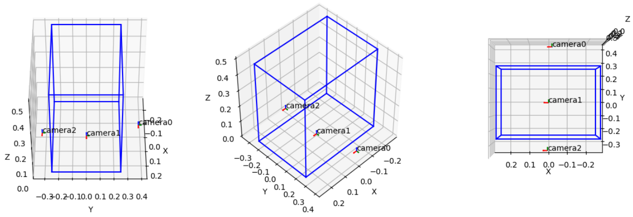


Figure 13: Extrinsic calibration results with camera positions and orientations.

Aim: Find the differences in calibration parameters in air and water.
Expected Outcome: Get the correction factor of calibration parameters from air to water.
Group member in charge: XU, Borong
Work: We repeated intrinsic calibration underwater and discovered that the calibration results failed in underwater environments. This could be due to light refraction when crossing different media.
To test the effectiveness of the intrinsic calibration results, we use the extrinsic calibration results to compare the localization between different cameras. Since the extrinsic parameters consist of transformations between cameras, they are not affected by light refraction. Once the extrinsic parameters are confirmed effective in air, we can use them to validate the intrinsic parameters produced by our method.
The cross-medium light refraction invalidates the pinhole camera assumption, and the traditional polynomial distortion model is insufficient to estimate the distortion. Therefore, we propose a new method for underwater distortion calibration.

Our new method calibrates the camera parameter K and distortion model D separately. We model the distortion parameter D as a vector field. We first apply the OpenCV calibration program with the assumption of no distortion to obtain the intrinsic parameter K, and using this K matrix, we normalize the object points and use the undistorted reprojected image points as ground truth for the distortion parameters. Then, we calibrate the distortion parameters using gradient descent with PyTorch. Using the new distortion model, we undistort the image points and recalibrate the intrinsic parameter K.

To reduce the effect of overfitting on the model training, we apply a train validation split of 8:2, and use the model with best performance on the validation set as our final model of distortion. For the model of the vector field, we apply a traditional polynomial distortion with radial and tangential components (Equation 3), as well as a Multi-Layer Perceptron (MLP) model with tanh activation function to model the nonlinear relationship. The learned distortion is visualized as a vector field (Figure 14), showing the per-pixel displacement applied to correct for underwater refraction. The different calibration methods are evaluated by cross-camera localization error using ground-truth extrinsic parameters, as shown in Figure 15.

$$
\begin{aligned}
r^2 &= x^2 + y^2 \\
x' &= x\underbrace{(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)}_{\text{radial}} + \underbrace{2p_1 xy + p_2(r^2 + 2x^2)}_{\text{tangential}} \\
y' &= y\underbrace{(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)}_{\text{radial}} + \underbrace{p_1(r^2 + 2y^2) + 2p_2 xy}_{\text{tangential}}
\end{aligned}
\tag{3}
$$

Algorithm 1 Separate K and Distortion Calibration

Require: Object points $\{\mathbf{Q}_i\}$, observed image points $\{\mathbf{p}_i\}$, image size
Ensure: Camera matrix K, distortion model D
  Step 1: Initial K calibration
  $K, \{R_j, \mathbf{t}_j\} \leftarrow \texttt{calibrateCamera}(\{\mathbf{Q}_i\}, \{\mathbf{p}_i\}, \text{dist} = \mathbf{0})$
  Step 2: Compute reprojected ground truth
  for each view j **do**
    $\hat{\mathbf{p}}_j \leftarrow K[R_j \mid \mathbf{t}_j]\,\mathbf{Q}_j$ ▷ Undistorted ground truth
  end for
  Step 3: Train distortion model
  Normalize: $\mathbf{x}_d \leftarrow K^{-1}\mathbf{p}$, $\quad \mathbf{x}_t \leftarrow K^{-1}\hat{\mathbf{p}}$
  Split $\{(\mathbf{x}_d, \mathbf{x}_t)\}$ into $\mathcal{T}_{\text{train}}$ (80%) and $\mathcal{T}_{\text{val}}$ (20%)
  Initialize distortion model D_θ
  for epoch $= 1$ **to** N_{epochs} **do**
    $\mathcal{L}_{\text{train}} \leftarrow \frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{(\mathbf{x}_d, \mathbf{x}_t) \in \mathcal{T}_{\text{train}}} \|D_\theta(\mathbf{x}_d) - \mathbf{x}_t\|^2$
    $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{train}}$
    $\mathcal{L}_{\text{val}} \leftarrow \frac{1}{|\mathcal{T}_{\text{val}}|} \sum_{(\mathbf{x}_d, \mathbf{x}_t) \in \mathcal{T}_{\text{val}}} \|D_\theta(\mathbf{x}_d) - \mathbf{x}_t\|^2$
    if $\mathcal{L}_{\text{val}} < \mathcal{L}^*_{\text{val}}$ **then**
      $\theta^* \leftarrow \theta$ ▷ Keep best model
    end if
  end for
  $D \leftarrow D_{\theta^*}$

2.2.3 Objective Statement 3: ArUco Tracking Implementation

The objective is to develop a tracking program based on ArUco marks for AUVs, controlling positional accuracy within $\pm 7 \times 10^{-2}$ m and achieving a detection speed of at least 10Hz.

Tasks:

- **Task 3.1: ArUco Localization Algorithm**

  Aim: Track ArUco markers of the AUVs.
  Expected Outcome: All the AUVs in the water tank can be tracked.
  Group member in charge: Both members
  Work: We wrote a Python program to implement real-time ArUco marker detection using OpenCV's ArUco module with the DICT_5X5_50 dictionary. Robust localization was achieved under varying lighting conditions. When capturing chessboard photos, we also recorded frames of an ArUco marker appearing at five fixed points and moving horizontally along a track (Figure 16) to test the algorithm. The algorithm visualized the coordinate axes of the marker and output the frame points associated with it.

  Before recording those frames, we planned the range of marker locations to be 2 m with 0.5 m between each fixed point. However, the table in the laboratory was not large enough for cameras to capture the marker across such a range. Therefore, the range was narrowed to 0.5 m with 0.125 m between each fixed point. We did not have any rails to move the marker in a straight line, but we found two straight grooves next to the table. We placed the tablet displaying the marker into the groove, which saved time and unnecessary expense.

- **Task 3.2: Multi-Camera Fusion**

  Aim: Combine detections of the ArUco markers from multiple cameras.
  Expected Outcome: The ArUco markers can be located according to detections from multiple cameras.
  Group member in charge: XU, Borong
  Work: We wrote an algorithm to combine detections from multiple cameras. Weighted averaging was imple-

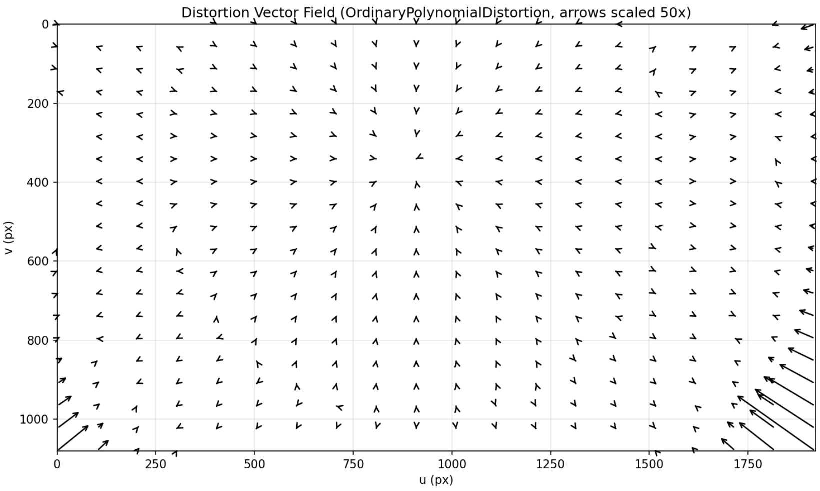


Figure 14: Vector field of the camera distortion.

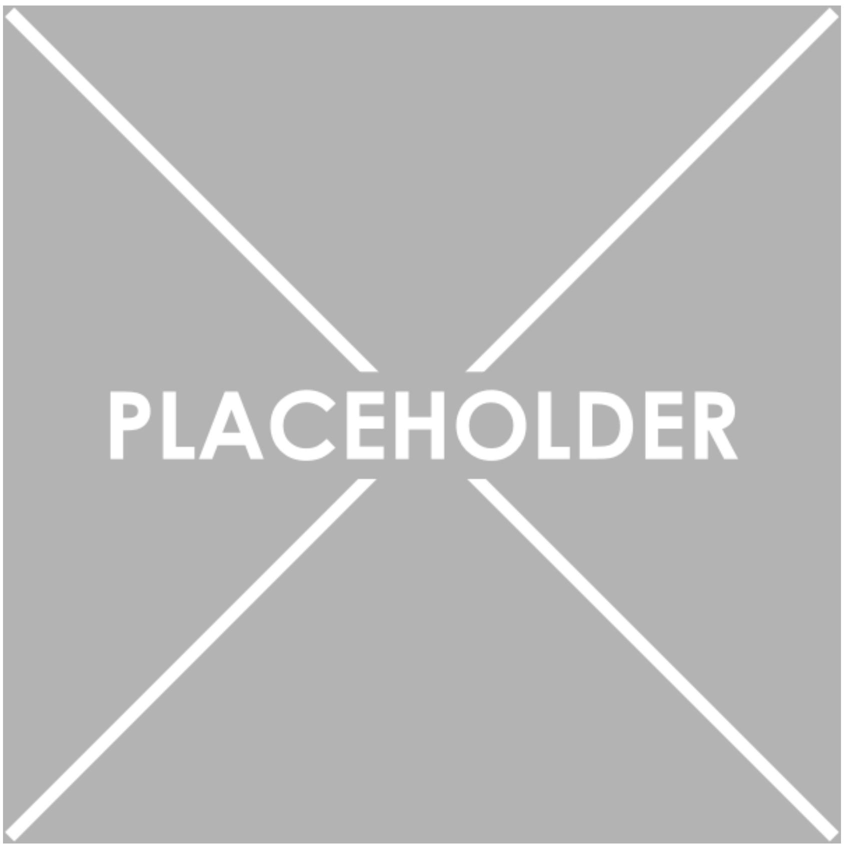


Figure 15: Comparison of intrinsic calibration methods.

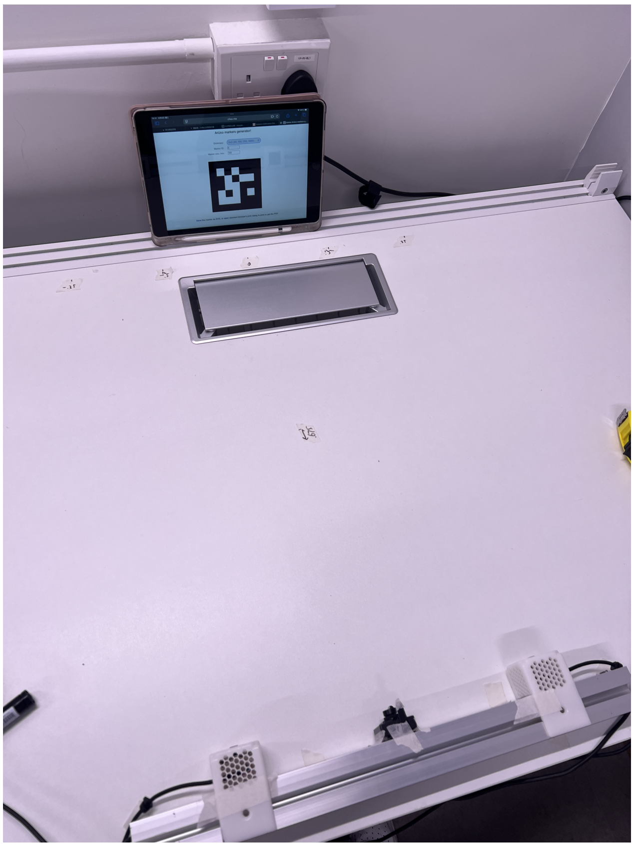


Figure 16: ArUco marker tracking test setup with marker moving along a linear path.

mented based on detection confidence and viewing angle. The algorithm also handles occlusions and marker visibility transitions between cameras by estimating the AUV's position from previous frames.

Figure 17 shows the position estimates from multiple cameras in the xy-plane, highlighting inconsistencies between camera coordinate systems before extrinsic calibration and fusion. After calibration, the in-air tracking results (Figure 18) show a smooth trajectory with minimal scatter, demonstrating accurate pose estimation without refractive distortion effects. In contrast, the underwater tracking results (Figure 19) using the same air calibration parameters show increased localization variability due to refraction effects, highlighting the need for refraction-corrected calibration parameters.

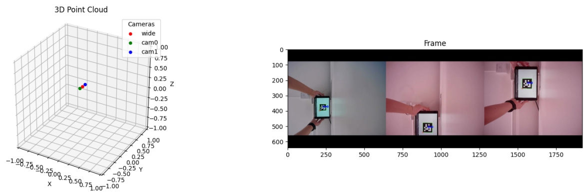
2.2.4 Objective Statement 4: RSSI localization Implementation

The objective is to utilize RSSI localization to get the coordinates of the AUVs for comparing with the results from optical localization.

Tasks:

- **Task 4.1: RSSI Fingerprints Mapping**

  Aim: Map the received signal strengths to the 3D distances between the RSSI sensor and an AUV.
  Expected Outcome: Error has to be within 7.0×10^{-2}m.
  Group member in charge: CHAN, Sze Sen
  Work: We wrote an algorithm in Arduino IDE to get the received signal strengths from the AUV and 4 fixed Wi-Fi access points (APs) nearby using an ESP32 Devkit S3 module board. In the experiment, the board was fixed at a



Camera	X	Y	Z
wide	-0.0353	0.0189	0.2204
cam0	-0.0354	-0.0949	0.2281
cam1	-0.0353	0.1275	0.2240

Figure 17: Multi-camera localization results of the same ArUco marker in the xy-plane.

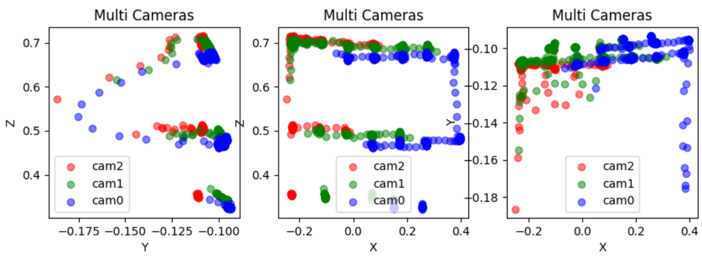


Figure 18: ArUco marker trajectory tracking results in air.

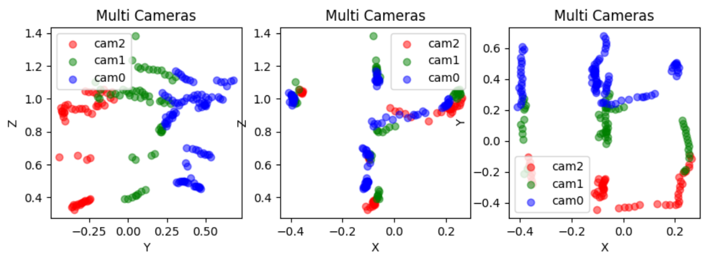


Figure 19: ArUco marker trajectory tracking results underwater using air calibration parameters.

corner of the testing area with coordinates (0,0,0). Fingerprints were taken at 0.1m spacing within 1.5m × 1.5m testing area by recording the mean signal strengths from the moving AUV and the fixed APs by the board. At each grid point, 100 samples were taken, and the mean of those samples was recorded. After that, the following vector was recorded as in Table 2: [x, y, z, AUV_RSSI, AP1_RSSI, AP2_RSSI, AP3_RSSI, AP4_RSSI]. These data were stored in a CSV file by the LittleFS filesystem.

When we were recording the received signal strengths, we found the fluctuation was high due to multipath effects and interference from the human body. To reduce the fluctuation, we increased the number of samples at each grid point and recorded the mean in the fingerprints. Another challenge we faced was preserving the data. Hundreds of fingerprints were recorded while the flash memory of the ESP32 board is limited. Addressing this challenge, we optimized the CSV format with only numbers and one type of symbol and monitored the file size periodically.

Table 2: Sample rows from the generated 3D fingerprint database.

x (m)	y (m)	z (m)	AUV RSSI	AP1 RSSI	AP2 RSSI	AP3 RSSI	AP4 RSSI
0.5	0.8	0.1	-45.2	-58.1	-67.3	-71.4	-75.3
1.2	0.3	0.5	-52.7	-49.8	-62.5	-68.9	-76.5

- **Task 4.2: AUV localization with RSSI**

Aim: Get locations of AUVs by matching mapped fingerprints.
Expected Outcome: All the AUVs in the water tank can be tracked.
Group member in charge: CHAN, Sze Sen
Work: We wrote an algorithm in Arduino IDE to estimate the 3D position of AUVs in real-time by matching live RSSI measurements against the pre-built fingerprint database from the last task. The localization was performed using Weighted K-Nearest Neighbors (WKNN) regression.

Let **x** be the live RSSI feature vector, and $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^{N}$ be the fingerprint database where $\mathbf{y}_i = (x_i, y_i, z_i)$ is the known 3D position.

1. **Distance Calculation (Euclidean distance in RSSI space):**

$$d_i = \|\mathbf{x} - \mathbf{x}_i\|_2 = \sqrt{\sum_{j=1}^{m}(x^{(j)} - x_i^{(j)})^2} \tag{4}$$

   where $m = 4$ (number of APs).

2. **Weight Calculation (Inverse Distance Weighting):**

$$w_i = \frac{1}{d_i + \epsilon}, \quad \text{subject to} \sum_{i=1}^{k} w_i = 1 \tag{5}$$

   where $\epsilon = 0.001$ is a small constant to avoid division by zero, and only the k nearest neighbors are considered.

3. **Position Estimation (Weighted Average):**

$$\hat{\mathbf{y}} = \sum_{i=1}^{k} w_i \cdot \mathbf{y}_i = \begin{bmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{bmatrix} \tag{6}$$

   where $k = 4$ was chosen after empirical tuning.

We tested the algorithm both in the air and underwater. However, we found that the difference in RSSI distribution between these environments is significant when using the same fingerprint map. Therefore, we repeated the last task to create a separate fingerprint map for the underwater environment.

2.3 Main Objective Evaluation & Discussion

The main objective of this project is to develop a fast and precise localization system for the Marine Automatic Swarm Experiment Platform (MASEP), enabling real-time tracking of AUVs with an error margin of $\pm 7 \times 10^{-2}$ m and a detection speed of at least 10 Hz. The system aims to provide a low-cost, scalable solution for multi-AUV tracking in swarm robotics research by combining multi-camera optical tracking with RSSI-assisted localization.

Summary of Results Table 3 summarizes the key results achieved for each objective statement.

Table 3: Summary of results for each objective statement.

Objective	**Target**	**Achieved**	**Status**
Simulation Environment	Working simulation with noise	Gamma noise model, EKF validated at 0.3×10^{-2} m	Completed
Intrinsic Calibration	$< 7.0 \times 10^{-2}$ m	7.1×10^{-2} m	Close to target
Extrinsic Calibration	$< 2.0 \times 10^{-2}$ m	2.2×10^{-2} m	Close to target
ArUco Tracking	$\pm 7.0 \times 10^{-2}$ m, $\geq$10 Hz	Functional in air; underwater needs improved calibration	Partially met
RSSI Localization	Sub-7.0×10^{-2} m	Signal strength mapping completed; localization in progress	In progress

Analysis of Results The simulation environment (Objective 1) was fully completed and validated. The Kalman filter fusion achieved 0.3×10^{-2} m error in simulation, well within the $\pm 7 \times 10^{-2}$ m target, confirming the viability of multi-camera fusion under idealized noise conditions.

For camera calibration (Objective 2), the intrinsic calibration error of 7.1×10^{-2} m is close to the 7.0×10^{-2} m target but has not fully met it. The extrinsic calibration error of 2.2×10^{-2} m slightly exceeds the 2.0×10^{-2} m target. Both results improved significantly from initial trials through increasing the number of valid captures from approximately 40 to 90 per camera. The primary challenge remains the cross-medium light refraction, which invalidates the standard pinhole camera assumption. Our proposed nonlinear distortion calibration method, which separately calibrates K and D using gradient descent with PyTorch, represents a novel approach to this problem. While the polynomial distortion model provides a baseline, the MLP-based model shows potential for better capturing the nonlinear refraction effects, though further tuning and data collection are required to fully validate its superiority.

For ArUco tracking (Objective 3), the system successfully detects and tracks markers in air with smooth trajectories. However, underwater tracking using air calibration parameters shows significantly increased variability (Figures 18 and 19), confirming that refraction-corrected calibration is essential for meeting the accuracy target underwater.

Comparison with Literature Compared to the acoustic localization methods reviewed in Section 1.3, which typically achieve 0.5–1 m accuracy [8, 9], our optical system provides approximately 10$\times$ better accuracy at significantly lower cost. The original MASEP localization system by Cai et al. [5] used only two fixed underwater cameras, whereas our five-camera array with overlapping views provides broader coverage and redundancy. Our ChArUco-based extrinsic calibration method addresses a key limitation not tackled in prior work: the inability of traditional chessboard calibration to function when the full pattern cannot be captured simultaneously by adjacent cameras underwater.

Compared to the RSSI localization methods by Naqvi et al. [13] (mean estimation error of 1.5 MSE) and Sun et al. [14] (0.12–0.49 m), our optical system achieves higher precision, though RSSI provides valuable complementary data in low-visibility conditions.

Assessment and Lessons Learned The project has successfully delivered a functional multi-camera localization system with two key technical contributions: the ChArUco-based extrinsic calibration algorithm and the nonlinear distortion calibration model. The simulation environment provided valuable insight into the effectiveness of Kalman filter fusion before real-world deployment.

The main challenge that prevented fully meeting the accuracy targets was the underwater refraction effect, which was more complex than initially anticipated. The traditional polynomial distortion model proved insufficient for cross-medium calibration, necessitating the development of the nonlinear approach. While the nonlinear model has shown promising results, additional training data and model refinement are needed to fully close the gap between air and underwater performance.

If the project were to be extended, collecting more underwater calibration data with diverse ChArUco board orientations and depths would likely improve the nonlinear distortion model accuracy. Additionally, integrating the RSSI localization data through the Extended Kalman Filter in real-world experiments (as validated in simulation) could provide the additional robustness needed to consistently meet the $\pm 7 \times 10^{-2}$ m target.

3 Conclusion

This project developed a multi-camera optical localization system for the Marine Automatic Swarm Experiment Platform (MASEP), targeting real-time AUV tracking with $\pm 7 \times 10^{-2}$ m accuracy. We successfully designed and implemented a complete pipeline from camera calibration through ArUco marker detection to multi-camera fusion.

Our main contributions are twofold. First, we proposed a ChArUco-based extrinsic calibration algorithm that addresses the fundamental limitation of traditional chessboard calibration in underwater environments. By leveraging the partial detection capability of ChArUco boards and PnP solvers, our method produces reliable inter-camera transformation matrices even when the full calibration pattern cannot be captured simultaneously by adjacent cameras. This achieved an extrinsic calibration error of 2.2×10^{-2} m. Second, we developed a nonlinear extrinsic calibration model that separately calibrates the camera intrinsic matrix K and distortion model D. By modeling distortion as a vector field and training it via gradient descent with PyTorch, this approach overcomes the cross-medium light refraction that invalidates the standard pinhole camera assumption and traditional polynomial distortion models.

The simulation environment with gamma-distributed noise modeling validated our Kalman filter fusion approach, achieving 0.3×10^{-2} m error under simulated conditions. The intrinsic calibration currently achieves 7.1×10^{-2} m error, which is close to but has not yet met our target. Further refinement of the nonlinear distortion model and additional calibration data collection are expected to reduce this error. RSSI localization was also investigated as an assistive method to improve robustness under challenging optical conditions such as low lighting and occlusion.

Overall, this project demonstrates that a low-cost multi-camera system with robust calibration techniques can provide accurate AUV localization for swarm robotics research, bridging the gap between simulation-based algorithm development and real-world underwater experiments.

Future work should focus on: (1) collecting additional underwater calibration data with diverse ChArUco board orientations to improve the nonlinear distortion model; (2) deploying the Extended Kalman Filter in real-world experiments to fuse optical and RSSI localization data, as validated in simulation; (3) conducting multi-AUV tracking experiments to evaluate system performance under realistic occlusion and inter-vehicle interference conditions; and (4) scaling the system to larger water tanks by increasing the number of cameras and optimizing the camera array geometry.

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Appendices

Appendix A - Project Schedule

Table 4: Project Gantt Chart (WK3–WK14)

Objective	Task	Member	WK3 11/10	WK4 18/10	WK5 25/10	WK6 1/11	WK7 8/11	WK8 15/11	WK9 22/11	WK10 29/11	WK11 6/12	WK12 13/12	WK13 20/12	WK14 27/12
Simulation Environment Development														
	Simulation Framework Setup	XU												
	Noise Modeling	XU												
	EKF Validation	XU												
Camera Array Calibration														
	Chessboard collection	CHAN												
	ChArUco collection	XU												
	Intrinsic Calibration	Both												
	Extrinsic Calibration	Both												
	Underwater Adjustment	Both												
	Nonlinear Distortion Calibration	XU												
ArUco Tracking														
	ArUco Localization	Both												
	Multi-Camera Fusion	XU												
RSSI Localization														
	RSSI Fingerprints Mapping	CHAN												
	Fingerprints Matching	CHAN												

Table 5: Project Gantt Chart (WK15–WK27)

Objective	Task	Member	WK15 3/1	WK16 10/1	WK17 17/1	WK18 24/1	WK19 31/1	WK20 7/2	WK21 14/2	WK22 21/2	WK23 28/2	WK24 7/3	WK25 14/3	WK26 21/3	WK27 28/3
Simulation Environment Development															
	Simulation Framework Setup	XU													
	Noise Modeling	XU													
	EKF Validation	XU													
Camera Array Calibration															
	Chessboard collection	CHAN													
	ChArUco collection	XU													
	Intrinsic Calibration	Both													
	Extrinsic Calibration	Both													
	Underwater Adjustment	Both													
	Nonlinear Distortion Calibration	XU													
ArUco Tracking															
	ArUco Localization	Both													
	Multi-Camera Fusion	XU													
RSSI Localization															
	RSSI Fingerprints Mapping	CHAN													
	Fingerprints Matching	CHAN													

Appendix B - Budget

Table 6: List of budget

Items	**Quantity**	**Cost per each (RMB¥)**
WXSJ-H65HD V01 camera	4	¥112
S-YUE 7730V1 camera	1	¥108
UGREEN CM475 USB-C adapter	1	¥169
3D printed Camera holder	4	¥0*
ESP32-S3-WROOM-1 module board	1	¥30
3D printed ESP32 holder	1	¥0*
M8 Screws 5pc	4	¥2.19
M8 T-Nut 10pc	2	¥1.6
M3 Screws and nuts 50pc	1	¥2.77
Aluminum extrusions 2020 1000mm	10	¥38.75
Aluminum extrusions 4040 1000mm	10	¥62.75
Calibration boards	5	¥0*
Contingency Budget (15%)		¥267.71
Total (including contingency)		¥2052.44

Budget Justification:

- **No-cost items (*):** 3D printed camera holders and calibration boards are provided by HKUST's Center for Engineering Education and Research Innovation (CEERI) through existing laboratory resources. This allocation supports academic research initiatives without additional expenditure, enabling access to professional-grade prototyping facilities.

- **Camera selection:** Five cameras (4× WXSJ-H65HD + 1× S-YUE) provide optimal cost-performance ratio at ¥110/camera average, offering 1280×960 resolution with OpenCV compatibility for real-time processing at 10 Hz.
- **Contingency allocation:** 15% contingency (¥267.71) covers potential hardware failures, additional calibration materials, or unexpected software licensing costs, ensuring project completion within budget constraints.
- **Cost optimization:** Total budget of ¥1784.74 represents significant savings compared to commercial AUV localization systems (typically ¥10,000+), achieving research-grade precision at educational price points while maintaining scalability for future multi-AUV experiments.

Appendix C - Meeting minutes

Meeting 1

- Date: 8/7/2025
- Time: 18:00
- Location: Zoom meeting
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
  - Prof. Huan YIN introduces the background information of this project
  - We start planning the main objectives
  - Prof. YIN also states some goals we have to achieve, and can challenge
  - Primary Goals:
    * Find the x-y coordinates of AUV
    * Find the z coordinates of AUV
    * Find the velocity of the AUV
  - Challenges:
    * Solve the location shift between cameras during localization
    * Localize the AUV with a single camera
    * Scan the AUV when part of the ArUco tag has been blocked

Table 7: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Visit CKSRI for setup inspection	13/7	CHAN, Sze Sen
Simulation Framework Setup	31/7	XU, Borong
Literature reviews on different localization methods	31/7	All

- Next Meeting: 5/8/2025, 19:30, Zoom meeting

Meeting 2

- Date: 5/8/2025
- Time: 19:30
- Location: Zoom meeting
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
  - We come up with all the main objectives of the project
  - We all agree to use 5 cameras for localization, and to try applying IMU odometry as a challenge part later for the project
  - We plan the experiment of camera calibration and ArUco markers localization
  - The experiment will be held on 6/8 and 14/8, CHAN, Sze Sen will go to CKSRI for the hardware part, while XU, Borong will work on the software part online
  - Prof. YIN suggests the settings for capturing images of ArUco markers localization

Table 8: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Visit CKSRI for setup inspection	13/7	CHAN, Sze Sen	Completed
Simulation Framework Setup	31/7	XU, Borong	Completed
Literature reviews on other localization methods	31/7	All	Completed

Table 9: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Image collection	14/8	CHAN, Sze Sen
Algorithms for intrinsic and extrinsic calibrations	14/8	XU, Borong
ArUco localization algorithm	22/8	All

- Next Meeting: 2/9/2025, 14:30, HKUST CKSRI

Meeting 3

- Date: 2/9/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
  - The errors of intrinsic and extrinsic calibrations are $\pm 7.1 \times 10^{-2}$ m and $\pm 2.2 \times 10^{-2}$ m
  - Prof. YIN suggests reduce the errors within $\pm 2 \times 10^{-2}$ m
  - We plan to continue the lab experiment after finishing the proposal report
  - Prof. YIN also suggests several papers that could be referred to in the proposal report

Table 10: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Image collection	14/8	CHAN, Sze Sen	Completed
Algorithms for intrinsic and extrinsic calibrations	14/8	XU, Borong	Completed
ArUco localization algorithm	22/8	All	Completed

Table 11: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Proposal Report Introduction	26/9	CHAN, Sze Sen
Proposal Report Methodology	26/9	XU, Borong
Finalize the proposal report	29/9	All

- Next Meeting: 8/10/2025, 14:30, HKUST CKSRI

Meeting 4

- Date: 8/10/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: XU, Borong
  - We completed calibration of the camera's intrinsic parameters for single-camera localization in air, designing and mounting the camera array under the experimental water tank
  - We also designed a new method for camera calibration using ChArUco boards to improve the accuracy of cross-camera views
  - We are now facing difficulties with the theoretical model of multi-camera calibration, which requires further literature review, and the gap between in-air and underwater cross-medium environments needs validation through underwater experiments
  - Prof. YIN provides feedback on refining the calibration approach, suggested additional references for modeling cross-medium distortion, and also suggested implying RSSI localization as an assisting method
  - We all agree to explore RSSI as an assisting localization method, model cross-medium light refraction, and conduct initial underwater experiments

Table 12: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Proposal Report Introduction	26/9	CHAN, Sze Sen	Completed
Proposal Report Methodology	26/9	XU, Borong	Completed
Finalize the proposal report	29/9	All	Completed

Table 13: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Literature review on RSSI for localization	22/10	CHAN, Sze Sen
Develop algorithm for cross-medium light refraction modeling	22/10	XU, Borong
Prepare and conduct initial underwater experiments	29/10	All

- Next Meeting: 5/11/2025, 14:30, HKUST CKSRI

Meeting 5

- Date: 5/11/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: XU, Borong
  - We completed calibration of the camera's intrinsic parameters for single-camera localization in both air and underwater, calibration of extrinsic parameters for the new camera array and mounting under the water tank
  - We also developed and implemented algorithms to localize ArUco marks
  - The literature review on the implementation of RSSI as an assisting localization method was conducted
  - We are now facing difficulties with underwater experiments, which reveal a gap between in-air and underwater intrinsic calibration, and a suitable Python environment for the RSSI system requires further research
  - Prof. YIN provides feedback on addressing the calibration gap through additional modeling and suggested references for RSSI implementation and Python environments
  - We all agree to investigate RSSI implementation through further literature reviews, use mathematical models to optimize non-linear point projection from 2D to 3D, conduct more underwater experiments, and develop a Python environment showing the relationship between signal strength and AUV position

Table 14: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Literature review on RSSI for localization	22/10	CHAN, Sze Sen	Completed
Develop algorithm for cross-medium light refraction modeling	22/10	XU, Borong	Completed
Prepare and conduct initial underwater experiments	29/10	All	Completed

Table 15: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Investigate implementation of RSSI through literature reviews	20/11	CHAN, Sze Sen
Use mathematical models to optimize non-linear point projection	20/11	XU, Borong
Conduct more underwater experiments to test the algorithm	27/11	All
Develop Python environment for signal strength and AUV position	27/11	All

- Next Meeting: 3/12/2025, 14:30, HKUST CKSRI

Meeting 6

- Date: 3/12/2025
- Time: 14:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
  - The development of the Python environment for signal strength and AUV position is still in progress since further investigation on the module board given by Prof. YIN is required
  - Prof. YIN mentions that he would leave HKUST, so the supervisor of this project would be changed to Prof. Fumin ZHANG
  - We plan to continue the lab experiment after finishing the progress report and midterm poster
  - Prof. YIN suggests that we should include more about the gaps of traditional localization methods and our plan for the remaining parts of this project in the progress report, and add more images instead of much text to draw attention in the midterm poster

Table 16: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Investigate implementation of RSSI through literature reviews	20/11	CHAN, Sze Sen	Completed
Use mathematical models to optimize non-linear point projection	20/11	XU, Borong	Completed
Conduct more underwater experiments to test the algorithm	27/11	All	Completed
Develop Python environment for signal strength and AUV position	27/11	All	In progress

Table 17: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Progress Report Objective Statement Execution	2/1	CHAN, Sze Sen
Progress Report Inroduction and Methodology Overview	2/1	XU, Borong
Midterm Postser	8/1	All
Finalize the Progress Report	9/1	All

- Next Meeting: 13/1/2026, 15:00, HKUST CKSRI

Meeting 7

- Date: 13/1/2026
- Time: 15:00
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
  - We discuss our division of work for the presentation in the midterm progress day on 23/1: CHAN, Sze Sen present the overview and objectives part; XU, Borong present the methodology and progress part
  - Prof. ZHANG suggests to use Arduino environment instead of Python for RSSI localization
  - We plan to test the reception of the signal strength of the ESP32 board with a simple algorithm in the Arduino environment, which only prints out the RSSI values of the target device
  - Prof. ZHANG also suggests that we can use a device such as a phone as a target device for reception testing

Table 18: Action Items from the Previous Meeting

Action Item to be completed	By when	By whom	Status
Progress Report Objective Statement Execution	2/1	CHAN, Sze Sen	Completed
Progress Report Inroduction and Methodology Overview	2/1	XU, Borong	Completed
Midterm Postser	8/1	All	Completed
Finalize the Progress Report	9/1	All	Completed

Table 19: Action Items for Next Meeting

Action Item to be completed	By when	By whom
Preparation of Midterm Progress Day	23/1	All
Testing of reception of the signal strength of the ESP32 board	2/2	XU, Borong
Develop Arduino environment for signal strength and AUV position	11/2	CHAN, Sze Sen

- Next Meeting: 11/2/2026, 14:00, HKUST CKSRI

Meeting 8

- Date: 11/2/2026
- Time: 15:00
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: XU, Borong
  - We designed an RSSI localization method and investigated the relationship between RSSI value and the distance between the target device and signal sensor
  - We also investigated the calibration algorithm in OpenCV to apply the non-linear calibration
  - We are now facing difficulties with significant errors in distance estimation, the signal from the target device is unstable, and the OpenCV calibration algorithm requires major modification for the non-linear calibration
  - Prof. ZHANG provides feedback on improving RSSI data collection setup and signal stability, and suggested further detailed analysis of OpenCV non-linear calibration algorithms
  - We all agree to increase the amount and range of signal data collection, conduct more reviews on the causes of signal instability, perform future investigations and detailed analyses of OpenCV camera calibration algorithms, and combine the non-linear calibration algorithm with the localization algorithm

Table 20: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Preparation of Midterm Progress Day	23/1	All	Completed
Testing of reception of the signal strength of the ESP32 board	2/2	XU, Borong	Completed
Develop Arduino environment for signal strength and AUV position	11/2	CHAN, Sze Sen	Completed

Table 21: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
Increase the amount and range of signal data collection	25/2	CHAN, Sze Sen
More review on the causes of signal instability	25/2	XU, Borong
Detailed analysis of OpenCV camera calibration algorithms	1/3	All
Combine the non-linear calibration algorithm with the localization algorithm	1/3	All

- Next Meeting: 24/3/2026, 16:30, HKUST CKSRI

Meeting 9

- Date: 24/3/2026
- Time: 16:30
- Location: HKUST CKSRI
- Attendees: CHAN, Sze Sen, XU, Borong
- Absent: None
- Minutes taken by: CHAN, Sze Sen
  - We plan to localize the AUVs by RSSI fingerprinting
  - Prof. ZHANG reminds us that we only get 1 ESP32 board, which is enough to localize the AUVs
  - We all agree to implement sensor fusion once the RSSI localization is completed
  - Prof. ZHANG also reminds us to catch up on our progress since it is near the end of the project

Table 22: Action Items from the Previous Meeting

Action Item to be completed	**By when**	**By whom**	**Status**
Increase the amount and range of signal data collection	25/2	CHAN, Sze Sen	Completed
More review on the causes of signal instability	25/2	XU, Borong	Completed
Detailed analysis of OpenCV camera calibration algorithms	1/3	All	Completed
Combine the non-linear calibration algorithm with the localization algorithm	1/3	All	Completed

Table 23: Action Items for Next Meeting

Action Item to be completed	**By when**	**By whom**
RSSI Fingerprinting	17/4	CHAN, Sze Sen
Final Report Objective Statement Execution update	17/4	CHAN, Sze Sen
Final Report Main Objective Evaluation & Discussion and Conclusion	17/4	XU, Borong
Final Poster	17/4	All
Implementation of sensor fusion	30/4	All

- Next Meeting: 21/4/2026, 16:30, HKUST CKSRI

Appendix D - Group Members' Contribution

Individual Contributions:

- **XU, Borong:** I led the algorithmic development of the localization system, covering the full pipeline from simulation to real-world deployment.

  For Objective 1, I designed and implemented the Python simulation environment using NumPy and Matplotlib, including virtual camera models with configurable intrinsic parameters, image point projection, and reprojection. I modeled the camera projection error as a gamma distribution for realistic noise simulation and implemented the Extended Kalman Filter for multi-camera fusion validation, achieving 0.3×10^{-2} m error in simulation.

  For Objective 2, I developed the core calibration algorithms. For intrinsic calibration, I wrote the algorithm that extracts focal length, principal point, and distortion coefficients from chessboard and ChArUco images. I proposed and implemented the ChArUco-based extrinsic calibration method, which is one of the main contributions of this project. This method uses PnP solvers to determine each camera's pose relative to the ChArUco board, enabling extrinsic calibration even when adjacent cameras cannot simultaneously capture the full calibration pattern. I also developed the nonlinear distortion calibration method that separately calibrates the camera matrix K and distortion model D using gradient descent with PyTorch, addressing the cross-medium refraction problem that invalidates conventional calibration approaches. I designed the camera array mounting structure and coordinated with CEERI for 3D printing of camera holders.

  For Objective 3, I implemented the multi-camera fusion algorithm with weighted averaging based on detection confidence and viewing angle, including occlusion handling and marker visibility transitions between cameras. I also contributed to the ArUco localization algorithm using OpenCV's ArUco module.

  For Objective 5, I implemented the Extended Kalman Filter for sensor fusion, combining results from different cameras with velocity estimation for the prediction step.

  I wrote the methodology sections of the proposal and progress reports, including all mathematical derivations and algorithm descriptions. I also took meeting minutes and participated in all laboratory experiments at HKUST CKSRI.

- **CHAN, Sze Sen:** I was responsible for hardware integration. To do this, I visited the HKUST CKSRI lab at the beginning of the project to inspect everything initially and ensure that the five-camera array was mounted correctly under the water tank we were using for our experiment, so that the overlapping fields of view from the cameras could provide optimal coverage.

  For Objective 1, I contributed to system specification and validation testing of the Python simulation environment for camera calibration.

  For Objective 2, I had to collect a large amount of data by capturing hundreds of chessboards and images in both air and underwater for camera calibration. I tried different pattern sizes and detection algorithms for the corners of these patterns to get the best results and minimize reprojection errors. For the experimental setup, I prepared and participated in initial and follow-up underwater tests, addressing challenges such as refraction distortions by validating air-water calibration gaps through hands-on trials and adjusting camera positions for better accuracy.

  For Objective 3, I also helped with validating the system by testing the ArUco marker detection in real-life scenarios. I found problems with how it worked on cameras and optimized the performance in order to achieve 10Hz detection speed.

  For Objective 4, I designed and implemented RSSI fingerprinting algorithm to localize multiple AUVs. I collected fingerprints of each grid point, including data of the coordinates of the point and the received signal strengths of the AUV and 4 Wi-Fi APs nearby from the ESP32 module board. I also localized the AUVs by matching the received signal strengths of the AUVs and the Wi-Fi APs with the taken fingerprints by the WKNN method.

  I prepared most of the meeting minutes and attached them to the report to fulfil the requirement. I prepared the introduction part of the proposal report. In this part, I stated the engineering problem of traditional localization methods. I also made literature reviews on existing solutions, including acoustic localization, magnetic localization, optical localization, and RSSI localization. In the progress report, I added details of the objective statement execution part, including images and stating the process and challenges of the tasks. For all reports, I have to ensure that they meet all the requirements of the guidelines, and the format of writing from both members is unified. I also designed the final poster by including the main points of the final report and the figures we obtained during the experiment.

  I coordinated laboratory resources by scheduling lab access, procuring components like cameras and adapters within budget, and ensuring proper experimental procedures, such as safety protocols during water tank operations.

  Throughout, because of the suggestion from Prof. YIN and Prof. ZHANG, I conducted literature reviews on alternative localization methods, including RSSI as an assisting technique, investigating its implementation through further readings and suggesting integrations to enhance optical tracking robustness under occlusions. This practical focus complemented the algorithmic work, bridging simulation to real-world validation.

Appendix E - Deviation from the proposal and progress report and supporting reason

The following deviations from the original proposal and progress report occurred during the project:

1. Addition of RSSI localization as an assistive method. The original proposal focused exclusively on optical localization. However, during laboratory experiments, we found that optical localization performance degraded significantly in low-light environments and under certain occlusion scenarios. Upon Prof. YIN's suggestion, we added RSSI localization as an assistive method to improve system robustness. This required additional hardware (ESP32 module board) and software development in the Arduino environment, which was not part of the original plan.

2. Transition from chessboard to ChArUco calibration boards. The proposal originally planned to use traditional chessboard patterns for all calibration tasks. During underwater experiments, we discovered that chessboard corner detection failed frequently due to complex light reflection disturbance underwater, and the requirement for full board visibility limited the range of angles available for extrinsic calibration. We deviated by adopting ChArUco boards, which allow partial detection and proved far more robust in underwater environments.

3. Development of nonlinear distortion calibration. The original plan assumed that standard OpenCV calibration (polynomial distortion model) would be sufficient for both air and underwater environments. Underwater experiments revealed that cross-medium light refraction invalidates the pinhole camera assumption, requiring us to develop a new calibration method that separately calibrates the camera matrix K and distortion model D using gradient descent. This was not anticipated in the proposal.

4. Sensor fusion implementation deferred. The proposal planned for full Extended Kalman Filter sensor fusion combining optical and RSSI data (Objective 5). While the EKF was implemented and validated in simulation, the real-world sensor fusion combining both modalities was not completed due to the additional time required for the nonlinear calibration development and RSSI system setup. The EKF was validated in the simulation environment with promising results.

Appendix F - Monthly Reports

Monthly Report for ECE FYP/FYT

<table>
<tr><td>Project Code:</td><td>YH02</td><td>Supervisor(s):</td><td>Yin Huan</td></tr>
<tr><td>Project Title:</td><td colspan="3">Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)</td></tr>
<tr><td>Group Member(s):</td><td colspan="3">1) CHAN, Sze Sen 2) XU, Borong</td></tr>
<tr><td>Reporting Period:</td><td colspan="3">Report #1 ☑ Oct (Fall)
Report #2 ☐ Nov (Fall)
Report #3 ☐ Feb (Spring)
(please attach Reports #1-2 to the Progress Report to be submitted in Jan)
(please attach Reports #3 to the Final Report to be submitted in Apr)</td></tr>
<tr><td>Progress Report:
● List the work completed in this reporting period.
● Identify the major difficulties encountered.
● Comment on the overall progress.</td><td colspan="3">Work completed:
● Calibrated the camera’s intrinsic parameter for single-camera localization in the air.
● Designed the camera array and mounted it under the experimental water tank.
● Designed a new method for camera calibration using ChArUco boards to improve the accuracy of cross-camera views.

Major difficulties:
● The theoretical model of multi-camera calibration is yet to be confirmed, requiring more literature review.
● The gap between an in-air environment and an underwater cross-medium environment is yet to be confirmed. Underwater experiments need to be conducted.

Overall progress:
● Current progress is matching our goals.</td></tr>
<tr><td>Future Plan:
● Write down the working plan for the next reporting period.</td><td colspan="3">● Discover the implementation of RSSI as an assisting localization method through literature reviews.
● Model the cross-medium light refraction and develop an algorithm to fit the distortion.
● Conduct underwater experiments.</td></tr>
<tr><td>Group Representative’s Signature:</td><td colspan="3"></td></tr>
</table>

(Version 2020-10)

Monthly Report for ECE FYP/FYT

Project Code:	YH02	Supervisor(s):	Yin Huan
Project Title:	Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)		
Group Member(s):	1) CHAN, Sze Sen 2) XU, Borong		
Reporting Period:	Report #1 ☑ Oct (Fall) Report #2 ☑ Nov (Fall) Report #3 ☐ Feb (Spring) (please attach Reports #1-2 to the Progress Report to be submitted in Jan) (please attach Reports #3 to the Final Report to be submitted in Apr)		
Progress Report: ● List the work completed in this reporting period. ● Identify the major difficulties encountered. ● Comment on the overall progress.	Work completed: ● Calibrated the camera's intrinsic parameters for single-camera localization in the air and under water. ● Calibrated the camera's extrinsic parameters for the new camera array and mount the camera array under the water tank. ● Developed and implemented algorithms to localize ArUco marks. ● Literature review on implementation of RSSI as an assisting localization method. Major difficulties: ● Underwater experiments revealed a gap between in-air and underwater intrinsic calibration. ● A suitable Python environment is yet to be confirmed for the RSSI system, requiring further research. Overall progress: ● Current progress is aligned with our goal, though slightly behind.		
Future Plan: ● Write down the working plan for the next reporting period.	● Investigate the implementation of RSSI as an assisting localization method through literature reviews. ● Use mathematical models to optimize the non-linear point projection from 2D point plane to 3D object point. ● Conduct more underwater experiments to test the algorithm. ● Develop a Python environment showing the relationship between signal strength and position of AUV		
Group Representative's Signature:	Borong Xu		

(Version 2020-10)

Monthly Report for ECE FYP/FYT

<table>
<tr><td>Project Code:</td><td>YH02</td><td>Supervisor(s):</td><td>Zhang Fumin</td></tr>
<tr><td>Project Title:</td><td colspan="3">Development of a Multi-Camera Vision-Based Localization System for Tracking Multiple Micro Autonomous Underwater Vehicles (AUVs)</td></tr>
<tr><td>Group Member(s):</td><td colspan="3">1) CHAN, Sze Sen 2) XU, Borong</td></tr>
<tr><td>Reporting Period:</td><td colspan="3">Report #1 ☐ Oct (Fall)
Report #2 ☐ Nov (Fall)
Report #3 ☑ Feb (Spring)
(please attach Reports #1-2 to the Progress Report to be submitted in Jan)
(please attach Reports #3 to the Final Report to be submitted in Apr)</td></tr>
<tr><td>Progress Report:

● List the work completed in this reporting period.
● Identify the major difficulties encountered.
● Comment on the overall progress.</td><td colspan="3">Work completed:
● Designed RSSI-based localization method.
● Investigated the relationship between RSSI value and distance between signal emitter and receiver.
● Investigate the calibration algorithm in OpenCV to apply the non-linear calibration.

Major difficulties:
● The distance estimation has significant errors. The setup of data collection has to be modified.
● Signal from the emitter is unstable. Experiment environment or choice of signal emitter has to be adjusted.
● The OpenCV calibration algorithm includes a backward mapping and an updated intrinsic parameter for localization, major modification should be made for the non-linear calibration algorithm.

Overall progress:
● Current progress is matching our goals.</td></tr>
<tr><td>Future Plan:

● Write down the working plan for the next reporting period.</td><td colspan="3">● Increase the amount and range of signal data collection.
● More review on the causes of signal instability.
● Future investigation and detailed analysis of OpenCV camera calibration algorithms.
● Combine the non-linear calibration algorithm with the localization algorithm.</td></tr>
<tr><td>Group Representative's Signature:</td><td colspan="3"></td></tr>
</table>

(Version 2020-10)